

Development Automation **& Linux Lab**

Experiment 6

Submitted By:

Ayush Singh Kushwah

SAP ID: 500095575

Enrolment No: R2142210216

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BTech CSE (DevOps)

Submitted To:

Mr. Durgesh Singh



SCHOOL OF COMPUTER SCIENCE
UNIVERSITY OF PETROLEUM & ENERGY STUDIES
Dehradun-248007
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EXPERIMENT-6

Automation Scripts to prevent Errors

1. Write a shell script to search and replace a given text input by the user with a new text across multiple files.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ cat ques1.sh
#!/bin/bash

read -p "Enter the text to search for: " SEARCH_TEXT
read -p "Enter the text to replace it with: " REPLACE_TEXT

FILES=$(find . -type f)

for FILE in $FILES; do
    grep -q "$SEARCH_TEXT" $FILE

    if [ $? -eq 0 ]; then
        sed -i "s/$SEARCH_TEXT/$REPLACE_TEXT/g" $FILE
        echo "Replaced \"$SEARCH_TEXT\" with \"$REPLACE_TEXT\" in $FILE"
    fi
done
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ cat a.txt
The moon is one of the most fascinating objects in the sky. It is a natural satellite that orbits around the Earth and can be seen from almost every part of the world. The moon is approximately one-quarter the size of the Earth and has a rocky, barren surface covered in craters, mountains, and valleys.

The moon's gravitational pull affects ocean tides, and its phases have been used to track time and calendar dates for centuries. The moon has also been the subject of many myths and legends across different cultures.

In recent years, space exploration has brought about new discoveries about the moon, including evidence of water on its surface. The moon also remains a destination for future space missions, with plans to establish a permanent human presence on the moon in the coming decades.

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ ./ques1.sh
Enter the text to search for: moon
Enter the text to replace it with: star
Replaced "moon" with "star" in ./a.txt
Replaced "moon" with "star" in ./b.txt
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ cat a.txt
The star is one of the most fascinating objects in the sky. It is a natural satellite that orbits around the Earth and can be seen from almost every part of the world. The star is approximately one-quarter the size of the Earth and has a rocky, barren surface covered in craters, mountains, and valleys.

The star's gravitational pull affects ocean tides, and its phases have been used to track time and calendar dates for centuries. The star has also been the subject of many myths and legends across different cultures.

In recent years, space exploration has brought about new discoveries about the star, including evidence of water on its surface. The star also remains a destination for future space missions, with plans to establish a permanent human presence on the star in the coming decades.
```

2. Write a shell script to check whether the given input is a valid Alphanumeric character.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ cat ques2.sh
#!/bin/bash

read -p "Enter a character: " char

if [[ "$char" =~ ^[[:alnum:]]+$ ]]; then
    echo "The input is a valid alphanumeric character."
else
    echo "The input is not a valid alphanumeric character."
fi
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ ./ques2.sh
Enter a character: 2
The input is a valid alphanumeric character.
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ ./ques2.sh
Enter a character: *
The input is not a valid alphanumeric character.
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ ./ques2.sh
Enter a character: 7
The input is a valid alphanumeric character.
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ ./ques2.sh
Enter a character: -
The input is not a valid alphanumeric character.
```

3. Write a shell script to transform the given input text to lowercase, uppercase and first letter uppercase.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ cat ques3.sh
#!/bin/bash

read -p "Enter some text: " text

echo "Lowercase: ${text,,}"
echo "Uppercase: ${text^^}"
echo "First letter uppercase: ${text^}"
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$ ./ques3.sh
Enter some text: ayush Singh KUSHWAH
Lowercase: ayush singh kushwah
Uppercase: AYUSH SINGH KUSHWAH
First letter uppercase: Ayush Singh KUSHWAH
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp6$
```